



AI Agent for GitHub Workflow Automation

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98–99%

Intent Recognition Accuracy

~99%

API Call Success Rate

Plain English → GitHub Actions

Hybrid NLP agent bridges the gap between non-technical stakeholders and GitHub workflows

~80%

LLM Dependency Reduction

7/7

Test Cases Passed

1. INTRODUCTION

GitHub is a cornerstone of modern software development, yet non-technical stakeholders; product managers, analysts, and coordinators; struggle to use it without developer help. This creates bottlenecks that delay project delivery.

This project presents a conversational AI Agent that translates plain English commands into GitHub API actions, enabling any team member to interact with GitHub without technical expertise. A **hybrid NLP architecture** combining rule-based logic with LLMs ensures both reliability and flexibility.

2. PROBLEM STATEMENT

Existing GitHub tools require technical knowledge of platform interfaces, command-line operations, or API usage; creating an accessibility gap for non-technical project stakeholders.

Goal: Design a conversational agent that maps natural language input to valid GitHub API actions, bridging the gap between plain English and structured software workflows.

3. OBJECTIVES

- Map natural language to GitHub actions (issues, PRs, repo activity)
- Integrate GitHub REST API v3 & GraphQL v4 for real-time operations
- Auto-generate daily/weekly repository activity & health reports
- Handle ambiguous or incomplete queries with intelligent disambiguation
- Reduce developer dependency for non-technical teams via autonomous execution

4. DATA SOURCES

Primary Source	GitHub REST API v3 with PAT Authentication
Secondary Source	GH Archive (open GitHub event logs)
Database	MongoDB (users, sessions, bugs, analytics)
Offline Dataset 1	commit_pr_dataset.csv (PushEvent, PullRequestEvent)
Offline Dataset 2	workflow_log.csv (WorkflowRunEvent, CI/CD logs)
Fields Captured	repo, commit_sha, author, message, date, status

Natural Language Input

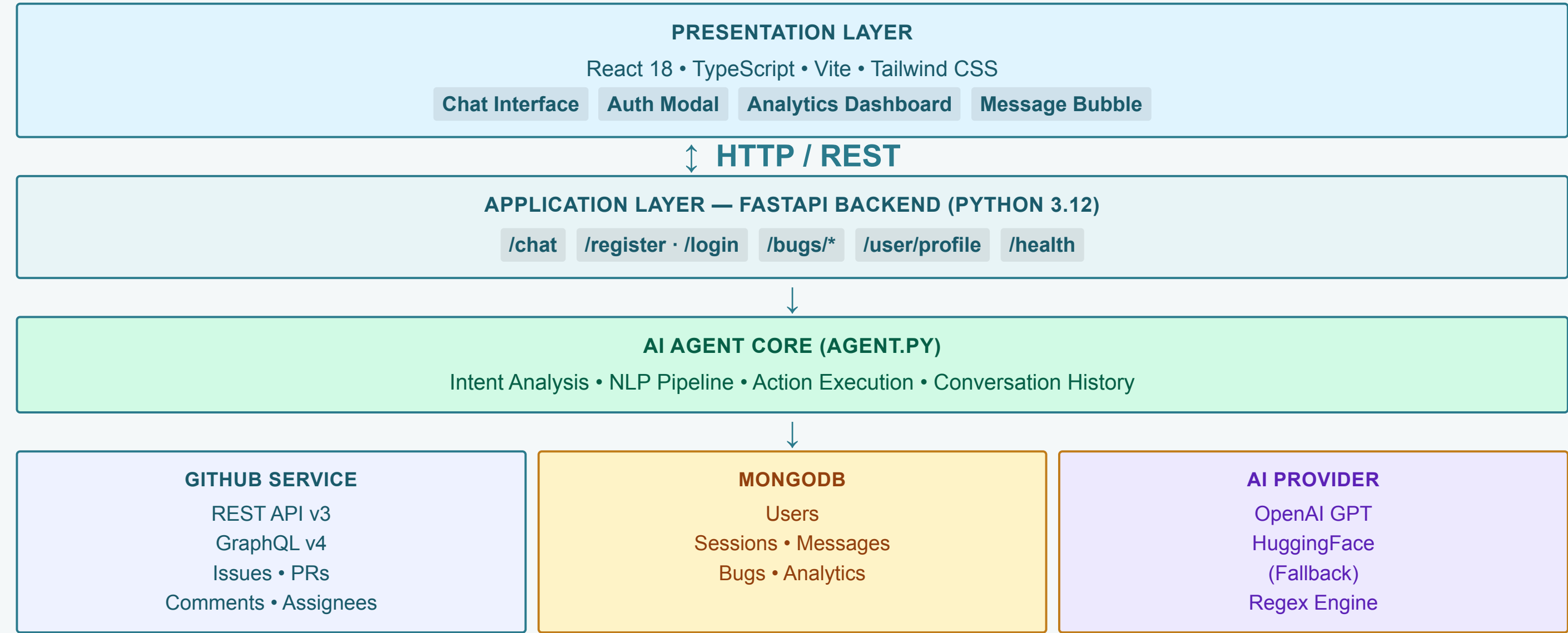
→ Agent Processing

→ GitHub Action

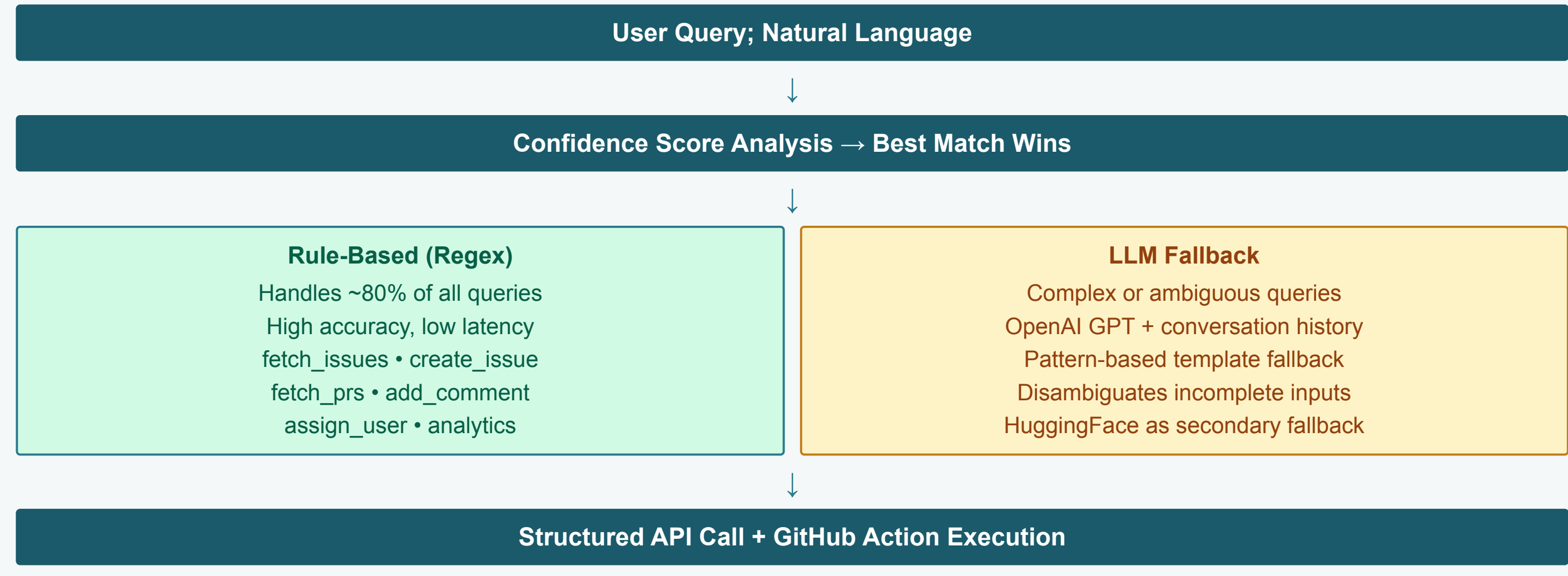
5. METHODOLOGY

User NL Input	Session Validation	Intent Analysis	Param Extraction	GitHub API Call	Response to User
Hybrid Routing Rule-based regex handles ~80% of queries; LLM invoked only for ambiguous inputs		LLM Fallback OpenAI GPT + HuggingFace for complex, multi-intent, or incomplete queries			
Session-Aware Full conversation history passed each turn for contextual disambiguation		Rate-Limit Safe 5-minute response cache reduces redundant API calls gracefully			
Deterministic Core Critical operations always use regex; zero LLM non-determinism risk		Real-Time Action Authenticated GitHub REST/GraphQL calls executed autonomously in milliseconds			

7. SYSTEM ARCHITECTURE

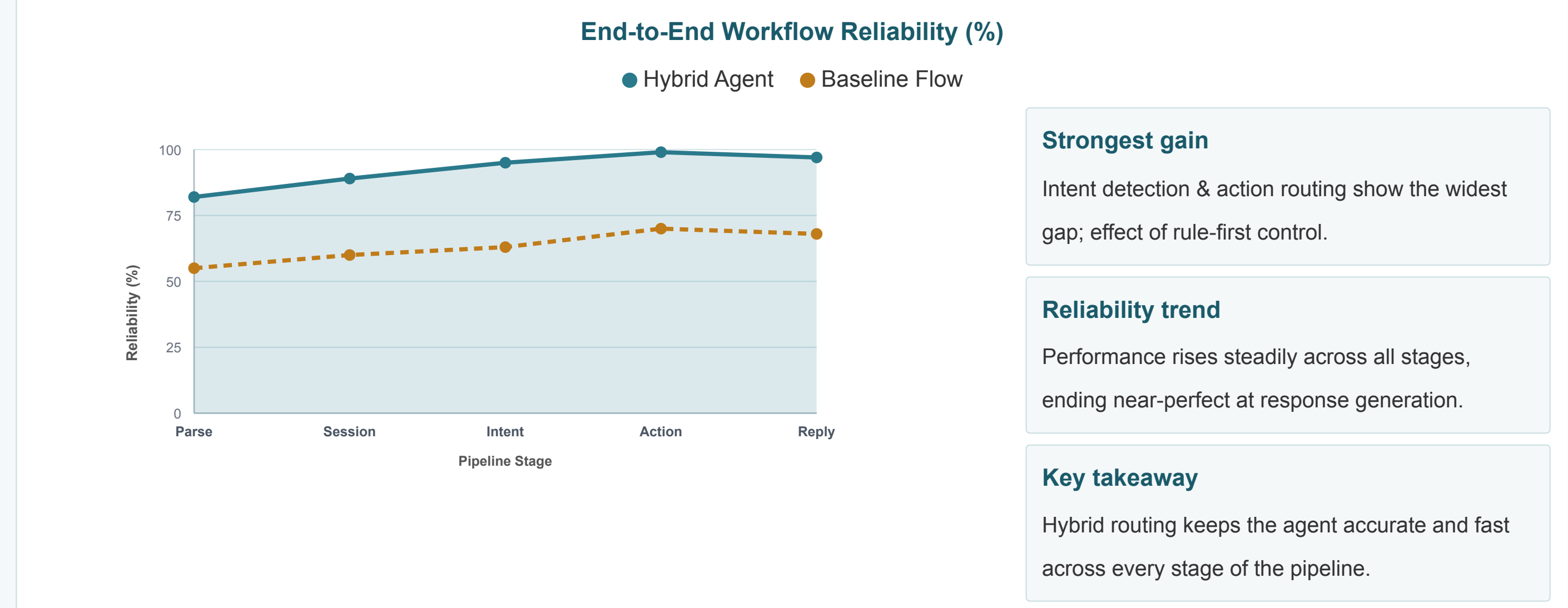


8. HYBRID INTENT DETECTION



9. RESULTS AND DISCUSSION

98–99% Intent Recognition Accuracy	~99% API Call Success Rate
97–98% Issue Creation Success Rate	~80% LLM Dependency Reduction



10. TESTING AND QUALITY ASSURANCE

Integration testing via Python pytest, Postman, and live GitHub REST API with PAT authentication.

ID	Feature Tested	Result
TC-001	Issue Creation: "Create a bug: Login button broken"	PASSED
TC-002	Issue Close: "Update issue #12: change status to closed"	PASSED
TC-003	Data Fetch: "Show all open issues in the repository"	PASSED
TC-004	Advanced Analytics: "Bugs opened in the last two weeks?"	PASSED
TC-005	Summarization: "Summarize this repo's weekly activity"	PASSED
TC-006	Error Handling: API rate limit exceeded gracefully	PASSED
TC-007	Disambiguation: Invalid or incomplete user input	PASSED

11. COMPARATIVE ANALYSIS

Feature	This System	GitHub Bots	General LLMs
Input Method	Natural Language	Slash Commands	Natural Language
Ease of Use	No tech knowledge	Technical required	Easy
Direct Action Execution	Autonomous	Predefined only	Manual steps needed
Built-in Analytics	Trend Reports	External service	Not available
Error Handling	Graceful Fallback	Raw API errors	No GitHub context
Task Execution Time	Low	High	High
LLM Dependency	Reduced 80%	None	Full dependency

12. KEY CHALLENGES & SOLUTIONS

Non-Deterministic LLM Output Fix: Hybrid system uses deterministic regex for all critical operations; LLM only for ambiguous queries.
GitHub API Rate Limits Fix: 5-minute caching for frequent data + graceful degradation with user-friendly error messages.
Structured Parameter Extraction Fix: Context-aware regex (e.g., issue #(\d+)) ensures accurate mapping from free text to typed API params.

13. FUTURE SCOPE

→ Transformer-based ML for enhanced intent classification	→ Multi-user & multi-repo enterprise-scale support
→ Slack & Teams integration for in-chat GitHub ops	→ Voice interface via speech-to-text for hands-free use
→ Autonomous multi-step agent with auto-assign & fix suggestions	→ Visual analytics dashboards for repo health reporting
→ CI/CD pipeline integration to monitor builds & trigger workflows	→ Role-based access control & encrypted token storage

14. REFERENCES

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